

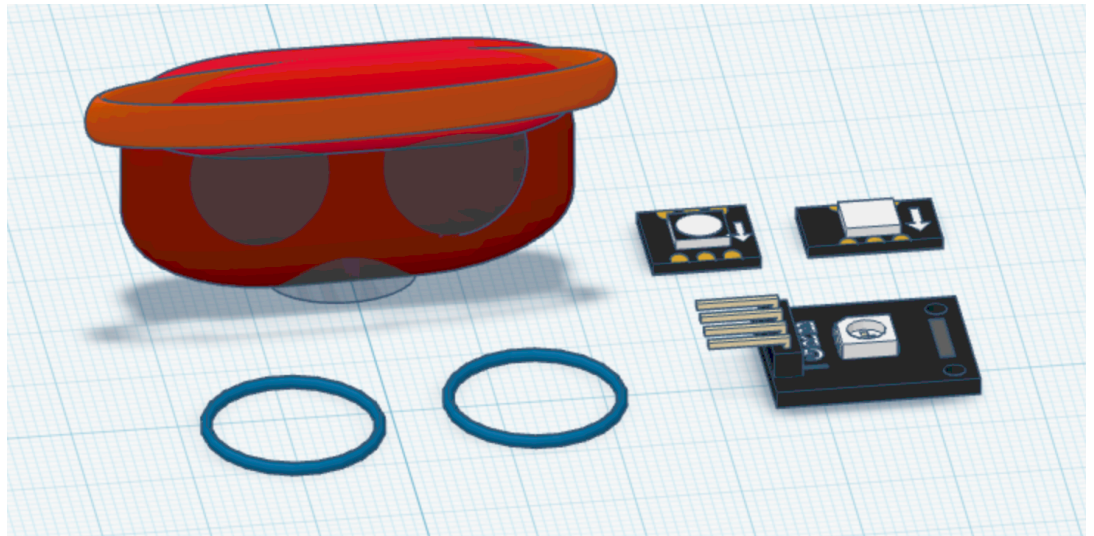


Project Title	Design and Development of a Circadian-Aligned IoT Sleep Mask for Sleep Data Analytics		
Course Title:	NCP4302 (NCP EL02)	Facilitator:	Engr. Errol John Antonnio
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Project Objectives:	The design project's goal is to develop a circadian-aligned IoT sleep mask that uses multi-modal stimuli like light and vibration to monitor sleep patterns, identify snoring patterns, and give an adaptive wake-up mechanism. Through a connected mobile application, it aims to increase wake-up efficiency, improve the quality of sleep, and give users useful sleep statistics.		
Functionalities:	The proposed IOT project aims to achieve the following functionalities: ■ Real-time monitoring of sleep duration using integrated sensors ■ Detection of snoring through sound-based sensing (microphone) ■ Automatic activation of flashing light when snoring is detected to prompt the user to adjust sleeping position ■ Detection of user movement and posture correction using a gyroscope sensor ■ Confirmation of position adjustment based on motion data from the gyroscope ■ Activation of a vibration module to assist in waking the user during light sleep or near wake-up state ■ Integration of a speaker module that serves as a sleep aid (e.g., calming sounds) and as an alarm output during scheduled wake-up time ■ Wireless data transmission via Wi-Fi to a mobile application ■ Data storage and synchronization using Firebase with user authentication ■ Visualization of sleep data, including sleep duration and snoring frequency, through a user dashboard ■ Adaptive and responsive wake-up mechanism based on detected sleep conditions		
Expected Prototype:	The expected prototype is a circadian-aligned wearable IoT sleep mask system developed using the Seed Xiao nRF52840 microcontroller as the central processing unit. The system integrates a microphone for acoustic-based snoring detection, a gyroscope sensor for monitoring body movement and sleep posture, an LED light module for visual stimulation, a vibration actuator for tactile feedback, and a speaker module for both sleep assistance and alarm generation. The system is designed to intelligently detect snoring events and initiate adaptive interventions through controlled light stimulation to prompt positional adjustment. Postural correction is subsequently validated through motion data acquired from the gyroscope sensor. Furthermore, the system incorporates a vibration-based mechanism to facilitate gradual and non-intrusive awakening during optimal wake-up periods, while the speaker module provides auditory sleep aids and scheduled alarm outputs. Wireless communication is established via Wi-Fi or Bluetooth, enabling seamless integration with a mobile application for real-time monitoring, alarm synchronization, and data visualization. Sleep-related data, including duration and snoring occurrences, are securely transmitted to and managed within a cloud-based platform utilizing authenticated user access. Overall, the prototype is expected to demonstrate a robust, adaptive, and user-centric IoT-based solution that leverages multi-modal feedback mechanisms and data-driven insights to enhance		



sleep quality and promote overall well-being.

Eye Sleep Mask with Sseed Xiao nRF52840, Accelerometer Gyro, Microphone Module with Amplifier, and 12-Bit RGB LED Ring.



Dashboard

